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Saeed Salehi

Associate Professor of Fluid Mechanics at Linköping University

Contact information

saeed.salehi@liu.se liu.se/en/employee/saesa86 saeedsalehi.com
Google Scholar ORCID ResearchGate salehisaeed
+46 72 018 87 28 Room 3A:970, A-Huset, Campus Valla, Linköping University, 581 83 Linköping, Sweden

Current Position

Associate Professor, *Linköping University* Aug. 2025 – present

Division of Applied Thermodynamics and Fluid Mechanics, Department of Management and Engineering. My research focuses on combining data-driven methods and artificial intelligence with Computational Fluid Dynamics (CFD) to develop efficient and reliable methods for simulation, model reduction, control, and uncertainty quantification of fluid flows.
I have also received my Docent title in Fluid Mechanics at Linköping University in February 2026.

Previous Appointments

Researcher, *Chalmers Industriteknik, based at Chalmers University of Technology* 2023 – 2025

Employed by Chalmers Industriteknik (CIT) and based full-time at Chalmers University of Technology, where the research was carried out. The work concerned the application of artificial intelligence and machine learning in computational fluid dynamics, within the project "Artificial Intelligence in Fluid Dynamics for Improving the Lifetime of Hydraulic Turbines".

Postdoctoral Researcher, *Chalmers University of Technology* 2019 – 2023

Project: New operating procedures of hydropower for a sustainable energy system. During my postdoctoral position, my research focused on developing advanced numerical methods in OpenFOAM to simulate the transient operations of hydraulic turbines. I also expanded my expertise in data-driven and machine-learning approaches, particularly in the field of reduced-order modeling, such as Proper Orthogonal Decomposition (POD) and Dynamic Mode Decomposition (DMD). The research resulted in several publications in high-impact, peer-reviewed journals and an open-source software package for simulating hydraulic turbine transients, which is being utilized by Vattenfall, a leading energy company in Sweden.

Lecturer, *University of Science and Culture, Tehran, Iran* 2018 – 2019

Taught "Introduction to Turbomachinery" at the undergraduate level.

Education and Academic Qualifications

Docent, *Linköping University, Fluid Mechanics* Feb. 2026

Qualifies for principal supervision of doctoral students; awarded on review of scientific and pedagogical qualifications.

PhD, *University of Tehran, Mechanical Engineering, Hydraulic Machinery* 2012 – 2018

Thesis: Uncertainty quantification of turbulent flows in hydraulic machinery
Supervisors: Prof. Mehrdad Raisee, Prof. Michel Cervantes, Prof. Ahmad Nourbakhsh
The thesis investigates the impact of operational and geometrical uncertainties on the performance of hydraulic machinery, extending uncertainty quantification (UQ) to this critical field using polynomial chaos expansion (PCE). To address the high computational costs of traditional UQ methods, I developed efficient techniques such as compressed sensing and multifidelity ℓ_1 minimization, validated through both analytical and CFD problems. Real-world case studies demonstrate that combined operational and geometrical uncertainties significantly influence the flow and performance of hydraulic pumps. This work advances UQ methods, contributing to more robust and reliable hydraulic machinery design.

MSc., *University of Tehran, Mechanical Engineering, Energy Conversion* 2010 – 2012

Thesis: Computation of steady and pulsating turbulent flow through a straight asymmetric diffuser with moderate adverse pressure gradient
Supervisors: Prof. Mehrdad Raisee, Prof. Michel Cervantes

BSc., *University of Tehran, Mechanical Engineering* 2006 – 2010

Thesis: Large eddy simulation of stall development around airfoils

Research Grants

Enabling Flexible Hydropower Operation by AI-Based Active Flow Control

Funding agency ÅForsk Foundation
Time period 2027 – 2031 (5 years)
Amount 4 000 000 SEK
Role **PI**

Awarded by the ÅForsk Foundation through their Early-Career Research Grant program. The project combines CFD, reinforcement learning-based active flow control, and hydropower to enable more flexible hydropower operation. A PhD student will be recruited at Linköping University. The total project budget including departmental co-funding is 5 550 000 SEK.

Artificial Intelligence for Enhanced Hydraulic Turbine Lifetime

Funding agency Swedish Hydropower Centre (SVC)
Time period 2023-01 – 2027-06 (four and a half years)
Amount 7 411 000 SEK
Role Co-PI (PI: Prof. Håkan Nilsson)

Developed the project concept and led the grant application. The award funded a researcher position (Co-PI) at Chalmers Industriteknik and Chalmers University of Technology. University in-kind contribution of 39.5% brings the total budget to 12 498 000 SEK, including a PhD student position.

Hydropower Operation and Lifetime Analysis

Funding agency Swedish Hydropower Centre (SVC)
Time period 2023-03 – 2027-08 (four and a half years)
Amount 5 266 000 SEK
Role Co-PI (PI: Prof. Håkan Nilsson)

Co-developed the application as a continuation of my postdoc research. The grant funded a PhD student (Martina Nobilo) in CFD for hydropower lifetime analysis, whom I co-supervised.

Multi-Fidelity Physics-Informed Neural Network to Solve Partial Differential Equations

Funding agency Chalmers University of Technology (internal, in competition)
Time period 2023-01 – 2027-06 (four and a half years)
Amount 5 517 000 SEK
Role Co-PI (collaborative application, no single main applicant)

This is the funding that we received for a cross-divisional PhD student in the Department of Mechanics and Maritime Sciences of Chalmers University of Technology. I played a pivotal role in formulating the concept and preparing the grant proposal application. Subsequently, the department approved our proposal, leading to the recruitment of a PhD student (Mohammad), whom I am co-supervising.

Publications

Peer-reviewed journal papers

The lists below are ordered by publication year, most recent first.

[1] Mohammad Sheikholeslami, **Saeed Salehi**, Wengang Mao, Arash Eslamdoost, and Håkan Nilsson. "State-of-the-art implementations of PINNs for the lid-driven cavity problem – a critical review with future perspectives". *Results in Engineering* 32 (2026), p. 112399. DOI: 10.1016/j.rineng.2026.112399.

[2] Martina Nobilo, **Saeed Salehi**, and Håkan Nilsson. "Lifetime analysis of hydro turbines with focus on fatigue damage in a renewable energy system – A review". *Renewable and Sustainable Energy Reviews* 228 (2026), p. 116578. DOI: 10.1016/j.rser.2025.116578.

[3] **Saeed Salehi**. "Transfer learning strategies for accelerating reinforcement-learning-based flow control" (2025). arXiv: 2510.16016 [cs.LG].

[4] Mohammad Sheikholeslami, **Saeed Salehi**, Wengang Mao, Arash Eslamdoost, and Håkan Nilsson. "Physics-informed neural networks with hard and soft boundary conditions for linear free surface waves". *Physics of Fluids* 37.8 (2025), p. 087158. DOI: 10.1063/5.0277421.

[5] **Saeed Salehi**. "An efficient intrusive deep reinforcement learning framework for OpenFOAM". *Meccanica* 60.6 (2025), pp. 1673–1693. DOI: 10.1007/s11012-024-01830-1.

[6] Jonathan Fahlbeck, Håkan Nilsson, Mohammad Hossein Arabnejad, and **Saeed Salehi**. "Performance characteristics of a contra-rotating pump-turbine in turbine and pump modes under cavitating flow conditions". *Renewable Energy* 237 (2024), p. 121605. DOI: 10.1016/j.renene.2024.121605.

[7] Faiz Azhar Masoodi, **Saeed Salehi**, and Rahul Goyal. "Reorganization of flow field due to load rejection driven self-mitigation of high load vortex breakdown in a Francis turbine". *Physics of Fluids* 36.9 (2024), p. 094110. DOI: 10.1063/5.0222739.

[8] **Saeed Salehi** and Håkan Nilsson. "Modal analysis of vortex rope using dynamic mode decomposition". *Physics of Fluids* 36.2 (2024), p. 024122. DOI: 10.1063/5.0186871.

[9] Faiz Azhar Masoodi, **Saeed Salehi**, and Rahul Goyal. "Formation and evolution of vortex breakdown consequent to post design flow increase in a Francis turbine". *Physics of Fluids* 36.2 (2024), p. 025116. DOI: 10.1063/5.0187104.

[10] **Saeed Salehi** and Håkan Nilsson. "A semi-implicit slip algorithm for mesh deformation in complex geometries, implemented in OpenFOAM". *Computer Physics Communications* 287 (2023), p. 108703. DOI: 10.1016/j.cpc.2023.108703.

[11] Jonathan Fahlbeck, Håkan Nilsson, and **Saeed Salehi**. "Surrogate based optimisation of a pump mode startup sequence for a contra-rotating pump-turbine using a genetic algorithm and computational fluid dynamics". *Journal of Energy Storage* 62 (2023), p. 106902. DOI: 10.1016/j.est.2023.106902.

[12] **Saeed Salehi** and Håkan Nilsson. "Effects of uncertainties in positioning of PIV plane on validation of CFD results of a high-head Francis turbine model". *Renewable Energy* 193 (2022), pp. 57–75. DOI: 10.1016/j.renene.2022.04.018.

[13] **Saeed Salehi** and Håkan Nilsson. "Flow-induced pulsations in Francis turbines during startup - A consequence of an intermittent energy system". *Renewable Energy* 188 (2022), pp. 1166–1183. DOI: 10.1016/j.renene.2022.01.111.

[14] Jonathan Fahlbeck, Håkan Nilsson, and **Saeed Salehi**. "A head loss pressure boundary condition for hydraulic systems". *OpenFOAM Journal* 2 (2022), pp. 1–12. DOI: 10.51560/ofj.v2.69.

[15] **Saeed Salehi**, Håkan Nilsson, Eric Lillberg, and Nicolas Edh. "An in-depth numerical analysis of transient flow field in a Francis turbine during shutdown". *Renewable Energy* 179 (2021), pp. 2322–2347. DOI: 10.1016/j.renene.2021.07.107.

[16] **Saeed Salehi** and Håkan Nilsson. "OpenFOAM for Francis turbine transients". *OpenFOAM Journal* 1 (2021), pp. 47–61. DOI: 10.51560/ofj.v1.26.

[17] Jonathan Fahlbeck, Håkan Nilsson, and **Saeed Salehi**. "Flow Characteristics of Preliminary Shutdown and Startup Sequences for a Model Counter-Rotating Pump-Turbine". *Energies* 14.12 (2021), p. 3593. DOI: 10.3390/en14123593.

[18] Mohamad Sadeq Karimi, Mehrdad Raisee, **Saeed Salehi**, Patrick Hendrick, and Ahmad Nourbakhsh. "Robust optimization of the NASA C3X gas turbine vane under uncertain operational conditions". *International Journal of Heat and Mass Transfer* 164 (2021), p. 120537. DOI: 10.1016/j.ijheatmasstransfer.2020.120537.

[19] Akbar Mohammadi Ahmar, **Saeed Salehi**, and Mehrdad Raisee. "Uncertainty quantification of the turbulent flow field and heat transfer of film cooling (in Persian)". *Journal of Solid and Fluid Mechanics* 10.2 (2020), pp. 177–192.

[20] Mohamad Sadeq Karimi, **Saeed Salehi**, Mehrdad Raisee, Patrick Hendrick, and Ahmad Nourbakhsh. "Probabilistic CFD computations of gas turbine vane under uncertain operational conditions". *Applied Thermal Engineering* 148 (2019), pp. 754–767. DOI: 10.1016/j.applthermaleng.2018.11.072.

[21] **Saeed Salehi**, Mehrdad Raisee, Michel J. Cervantes, and Ahmad Nourbakhsh. "On the flow field and performance of a centrifugal pump under operational and geometrical uncertainties". *Applied Mathematical Modelling* 61 (2018), pp. 540–560. DOI: 10.1016/j.apm.2018.05.008.

[22] **Saeed Salehi**, Mehrdad Raisee, Michel J. Cervantes, and Ahmad Nourbakhsh. "An efficient multifidelity ℓ_1-minimization method for sparse polynomial chaos". *Computer Methods in Applied Mechanics and Engineering* 334 (2018), pp. 183–207. DOI: 10.1016/j.cma.2018.01.055.

[23] Ali Salehpour, **Saeed Salehi**, Samaneh Salehpour, and Mehdi Ashjaee. "Thermal and hydrodynamic performances of MHD ferrofluid flow inside a porous channel". *Experimental Thermal and Fluid Science* 90 (2018), pp. 1–13. DOI: 10.1016/j.expthermflusci.2017.08.032.

[24] **Saeed Salehi**, Mehrdad Raisee, Michel J. Cervantes, and Ahmad Nourbakhsh. "Efficient uncertainty quantification of stochastic CFD problems using sparse polynomial chaos

and compressed sensing". *Computers & Fluids* 154 (2017), pp. 296–321. DOI: 10.1016/j.compfluid.2017.06.016.

[25] **Saeed Salehi**, Mehrdad Raisee, Michel J. Cervantes, and Ahmad Nourbakhsh. "The Effects of Inflow Uncertainties on the Characteristics of Developing Turbulent Flow in Rectangular Pipe and Asymmetric Diffuser". *Journal of Fluids Engineering* 139.4 (2017), p. 041402. DOI: 10.1115/1.4035302.

[26] **Saeed Salehi**, Mehrdad Raisee, and Michel J. Cervantes. "Computation of Developing Turbulent Flow Through a Straight Asymmetric Diffuser With Moderate Adverse Pressure Gradient". *Journal of Applied Fluid Mechanics* 10.4 (2017), pp. 1029–1043. DOI: 10.18869/acadpub.jafm.73.241.26311.

[27] **Saeed Salehi** and Mehrdad Raisee. "Application of Gram-Schmidt orthogonalization method in uncertainty quantification of computational fluid dynamics problems with arbitrary probability distribution functions (In Persian)". *Modares Mechanical Engineering* 15.12 (2015), pp. 1–8.

[28] Behrouz Takabi and **Saeed Salehi**. "Augmentation of the Heat Transfer Performance of a Sinusoidal Corrugated Enclosure by Employing Hybrid Nanofluid". *Advances in Mechanical Engineering* 6 (2014), p. 147059. DOI: 10.1155/2014/147059.

Book chapters

[1] **Saeed Salehi**, Mehrdad Raisee, Michel Cervantes, and Ahmad Nourbakhsh. "Development of an Efficient Multifidelity Non-Intrusive Uncertainty Quantification Method". *Evolutionary and Deterministic Methods for Design Optimization and Control With Applications to Industrial and Societal Problems*. Ed. by Esther Andrés-Pérez, Leo M. González, Jacques Periaux, Nicolas Gauger, Domenico Quagliarella, and Kyriakos Giannakoglou. Springer International Publishing, 2019, pp. 483–497. DOI: 10.1007/978-3-319-89890-2_31.

Conference proceedings

[1] **Saeed Salehi** and Håkan Nilsson. "A parametric study of axial flow jets for mitigation of vortex rope instabilities". *IOP Conference Series: Earth and Environmental Science*. Vol. 1561. 1. IOP Publishing, 2025, p. 012023. DOI: 10.1088/1755-1315/1561/1/012023.

[2] Martina Nobilo, **Saeed Salehi**, and Håkan Nilsson. "On the flow-induced pulsating forces during load reduction of a Kaplan turbine model". *IOP Conference Series: Earth and Environmental Science*. Vol. 1483. 1. IOP Publishing, 2025, p. 012022. DOI: 10.1088/1755-1315/1483/1/012022.

[3] J Fahlbeck, H Nilsson, and **Saeed Salehi**. "On the pump mode shutdown sequence for a model contra-rotating pump-turbine". *IOP Conference Series: Earth and Environmental Science*. Vol. 1483. 1. IOP Publishing, 2025, p. 012016. DOI: 10.1088/1755-1315/1483/1/012016.

[4] **Saeed Salehi** and Håkan Nilsson. "Dynamic Mode Decomposition of Rotating Vortex Rope Instability". *9th IAHR Meeting of the WorkGroup on Cavitation and Dynamic Problems in Hydraulic Machinery and Systems*. 2024.

[5] Martina Nobilo, **Saeed Salehi**, and Håkan Nilsson. "Effects of load reduction on forces and moments on the runner blades of a Kaplan turbine model". *IOP Conference Series: Earth and Environmental Science*. Vol. 1411. 1. IOP Publishing, 2024, p. 012001. DOI: 10.1088/1755-1315/1411/1/012001.

[6] J Fahlbeck, H Nilsson, and **Saeed Salehi**. "Analysis of mode-switching of a contra-rotating pump-turbine based on load gradient limiting shutdown and startup sequences". *IOP Conference Series: Earth and Environmental Science*. Vol. 1411. 1. IOP Publishing, 2024, p. 012049. DOI: 10.1088/1755-1315/1411/1/012049.

[7] Mohammad Sheikholeslami, **Saeed Salehi**, Wengang Mao, Arash Eslamdoost, and Håkan Nilsson. "Physics-Informed Neural Networks for Modeling Linear Waves". *International Conference on Offshore Mechanics and Arctic Engineering. Philip Liu Honoring Symposium on Water Wave Mechanics and Hydrodynamics; Blue Economy Symposium*. Vol. 9. 2024, V009T12A002. DOI: 10.1115/OMAE2024-125048.

[8] Jonathan Fahlbeck, Håkan Nilsson, and **Saeed Salehi**. "Evaluation of startup time for a model contra-rotating pump-turbine in pump-mode". *IOP Conference Series: Earth and Environmental Science*. Vol. 1079. 1. IOP Publishing, 2022, p. 012034. DOI: 10.1088/1755-1315/1079/1/012034.

[9] **Saeed Salehi**, Håkan Nilsson, Eric Lillberg, and Nicolas Edh. "Development of a novel numerical framework in OpenFOAM to simulate Kaplan turbine transients". *IOP Conference Series: Earth and Environmental Science*. Vol. 774. 1. IOP Publishing, 2021, p. 012058. DOI: 10.1088/1755-1315/774/1/012058.

[10] **Saeed Salehi**, Håkan Nilsson, Eric Lillberg, and Nicolas Edh. "Numerical Simulation of Hydraulic Turbine During Transient Operation Using OpenFOAM". *IOP Conference Series: Earth and Environmental Science*. Vol. 774. 1. IOP Publishing, 2021, p. 012060. DOI: 10.1088/1755-1315/774/1/012060.

[11] Jonathan Fahlbeck, Håkan Nilsson, **Saeed Salehi**, Mehrdad Zangeneh, and Melvin Joseph. "Numerical analysis of an initial design of a counter-rotating pump-turbine". *IOP Conference Series: Earth and Environmental Science*. Vol. 774. 1. IOP Publishing, 2021, p. 012066. DOI: 10.1088/1755-1315/774/1/012066.

[12] Mohamad Sadeq Karimi, **Saeed Salehi**, Mehrdad Raisee, and Ahmad Nourbakhsh. "Conjugate Heat Transfer Simulation of a Cooled Turbine Vane under Uncertain Operational Condition". *International Conference on Evolutionary and Deterministic Methods for Design Optimization and Control with Applications to Industrial and Societal Problems, Madrid, Spain*. 2017.

[13] **Saeed Salehi**, Mehrdad Raisee, Michel Cervantes, and Ahmad Nourbakhsh. "Development of an Efficient Multifidelity Non-Intrusive Uncertainty Quantification Method". *International Conference on Evolutionary and Deterministic Methods for Design Optimization and Control with Applications to Industrial and Societal Problems, Madrid, Spain*. 2017.

[14] **Saeed Salehi**, Mehrdad Raisee, and Ahmad Nourbakhsh. "Effects of Geometrical and Operational Uncertainties on the Performance of Hydraulic Machinery". *The 14th Asian International Conference on Fluid Machinery, Zhenjiang, China*. 2017.

[15] Vahid Etemadeasl, **Saeed Salehi**, Mehrdad Raisee, and Ahmad Nourbakhsh. "Numerical Investigation of Turbulent Flow in Francis-99 Turbine Using Various Turbulence Models". *The 14th Asian International Conference on Fluid Machinery, Zhenjiang, China*. 2017.

[16] Ehsan Akbari, **Saeed Salehi**, and M. Karami. "Numerical and Experimental Investigation of the Effect of Heat Exchanger Shape on Performance of an Industrial Heater (In Persian)". *21st Annual International Conference on Mechanical Engineering (ISME), Tehran, Iran*. 2013.

[17] Behrouz Takabi, **Saeed Salehi**, and Mohammad Hassan Rahimyan. "Studying the effects of employing hybrid nanofluid on heat transfer performance of a wavy cavity". *21st Annual International Conference on Mechanical Engineering (ISME), Tehran, Iran*. 2013.

Conference presentations

[1] **Saeed Salehi**. "Closed-loop control of vortex rope in a swirl generator using deep reinforcement learning". 2026. 21st OpenFOAM Workshop, Guimarães, Portugal (OFW21).

[2] **Saeed Salehi**. "Transfer learning strategies for accelerating reinforcement-learning-based flow control". 2026. 3rd ERCOFTAC Workshop on Machine Learning for Fluid Dynamics, CWI, Amsterdam, Netherlands.

[3] **Saeed Salehi**. "CFD with OpenSource Software: a case study of project-based learning in an international PhD course". 2024. Chalmers Conference on Teaching and Learning (KUL).

[4] **Saeed Salehi**. "Towards practical applications of deep reinforcement learning in CFD". 2024. ERCOFTAC Workshop on Machine Learning for Fluid Dynamics, Sorbonne University, Paris, France.

[5] **Saeed Salehi**. "Implementation of deep reinforcement learning in OpenFOAM for active flow control". 2023. 18th OpenFOAM Workshop, Genoa, Italy (OFW18).

[6] **Saeed Salehi**. "Deep reinforcement learning for active flow control". 2023. Dutch OpenFOAM User Group Meeting, July, TU Delft, Netherlands.

[7] **Saeed Salehi**. "Evolution of flow features during transient operation of a Kaplan turbine". 2022. 17th OpenFOAM Workshop, Cambridge, UK (OFW17).

[8] **Saeed Salehi**. "Numerical simulation of hydraulic turbine during transient operation using OpenFOAM". 2020. 15th OpenFOAM Workshop, Virginia, USA (OFW15).

[9] **Saeed Salehi**. "Quantification of epistemic uncertainties in the $k-\varepsilon$ model coefficients". 2019. Gothenburg Region OpenFOAM User Group Meeting (OFGBG19).

Supervision

I am supervising or have supervised the following students.

- **Jonathan Fahlbeck**, PhD student, Chalmers University of Technology, 2020 – 2024 (graduated), co-supervisor. *Project:* Flow in contra-rotating pump-turbines at stationary, transient, and cavitating conditions (ALPHEUS project – European Union's Horizon 2020 program)
- **Martina Nobilo**, PhD student, Chalmers University of Technology, 2023 – 2025 (graduated with Licentiate degree), co-supervisor. *Project:* CFD for hydropower lifetime analysis
- **Mohammad Sheikholeslami**, PhD student, Chalmers University of Technology, 2023 – present, co-supervisor. *Project:* Multi-Fidelity Physics-Informed Neural Network to solve partial differential equations

Teaching

This summary outlines my primary teaching experiences in academia, showcasing my contributions to undergraduate and graduate education. Fuller detail on my academic teaching, my teaching experience beyond academia, my supervision, and my final pedagogical project is given in my pedagogical portfolio, which is supplied as a separate document with my applications and is otherwise available on request.

- **Applied Computational Fluid Dynamics** (TMMV59, 6 credits) – Examiner, course coordinator, and main teacher. Linköping University, Master's level. 2026 – present.
- **Machine Learning for Mechanical Engineering** (TMMV64, 6 credits) – Examiner, course coordinator, and main teacher. Linköping University, Master's level. 2026 – present.
- **Introduction to fluid dynamics in water turbines** (2.0 ECTS) – Main teacher and examiner. SVC research school at KTH Royal Institute of Technology, PhD level, ∼5 students/year. 2022.
- **Basic Usage of OpenFOAM** (2.0 ECTS) – Lecturer, teaching assistant. Chalmers University of Technology, PhD level, 17–30 students/year. 2021 – present.
- **CFD with OpenSource Software** (7.5 ECTS) – Lecturer, teaching assistant. Chalmers University of Technology, PhD level, 7–13 students/year. 2020 – present.
- **Introduction to turbomachinery** (7.5 ECTS, converted) – Main teacher and examiner. University of Science and Culture, Iran, Bachelor level, ∼20 students/year. 2018 – 2019.
- **Teaching assistant** (Turbomachinery Lab., Turbulent Flows, Introduction to Heat Transfer, Introduction to Fluid Mechanics) – University of Tehran, Iran, Bachelor and Master level. 2013 – 2017.

Pedagogical training

In October 2023, I was awarded the "Diploma in Teaching and Learning in Higher Education" in recognition of the successful fulfillment of all program requirements, including seven courses. The Diploma is designed to meet the learning outcomes defined by the Association of Swedish Higher Education Institutions (SUHF). The program is equivalent to 15 ECTS credits or 10 weeks of full-time study. It concludes with conducting a final pedagogical project related to the teaching activities. More information about my final project is provided in my pedagogical portfolio.

Additionally, to further refine my abilities in the realm of supervision, I completed the "Supervising Research Students" course (3.0 ECTS) which is not included in the teaching diploma program. Together with the diploma program, this amounts to **18.0 ECTS** of pedagogical training completed at Chalmers University of Technology, complemented by a further **4 credits** completed at Linköping University in 2025. The courses are listed below.

- **Higher education pedagogy for PhD supervisors** (4 credits) – Linköping University, December 2025, including the faculty-specific part for the Faculty of Science and Engineering (Institute of Technology). The cross-university part was assessed as equivalent to *Supervising research students* below.

- **Diversity and inclusion for learning in higher education** (2.0 ECTS) – Chalmers University of Technology, 2023.

- **Supervising writing processes** (2.5 ECTS) – Chalmers University of Technology, 2023.

- **Theoretical perspectives on learning** (2.5 ECTS) – Chalmers University of Technology, 2023.

- **Enhancing learning through writing** (4.5 ECTS) – Chalmers University of Technology, 2023.

- **Minor independent study in teaching and learning in higher education** (0.5 ECTS) – Chalmers University of Technology, 2023.

- **Reflections on teaching and learning in higher education** (0.5 ECTS) – Chalmers University of Technology, 2023.

- **Supervising research students** (3.0 ECTS) – Chalmers University of Technology, 2022. Not part of the teaching diploma program.

- **University teaching and learning** (2.5 ECTS) – Chalmers University of Technology, 2021.

Academic Service

Editorial roles

Section Editor, OpenFOAM Journal, 2026 – present

Committees

Technical Committee Member, OpenFOAM Turbomachinery Technical Committee, OpenFOAM Governance, 2022 – present

Conference service

Session Chair, Multiphase Flows I session, 21st OpenFOAM Workshop (OFW21), Guimarães, Portugal, July 2026

Session Chair, Reinforcement Learning (RL3) session, 3rd ERCOFTAC Workshop on Machine Learning for Fluid Dynamics, CWI, Amsterdam, the Netherlands, March 2026

Doctoral examination and progress review

External Reviewer (90% seminar), University of Gävle, Sweden, 2026
Thesis: "Experimental and Numerical Evaluation of Wall-Confluent Jets Thermal Performance for Greenhouse Climate Control". PhD candidate: Gasper Choonya

External Examiner (Defense committee), Universitat Rovira i Virgili, Tarragona, Spain, 2024
Thesis: "Numerical Modeling of High Aspect Ratio Fibers in Fluid Flows". PhD candidate: MohammadJavad Norouzi

Reviewer for

Physics of Fluids (*AIP*), AIAA Journal (*AIAA*), Computers & Fluids (*Elsevier*), Applied Mathematical Modelling (*Elsevier*), Journal of Fluids Engineering (*ASME*), Biomechanics and Modeling in Mechanobiology (*Springer*), Science Progress (*Sage*), Fluids (*MDPI*), Energies (*MDPI*), Algorithms (*MDPI*), Processes (*MDPI*), Journal of Applied Fluid Mechanics (*IUT*)

Invited Talks and Expert Assignments

- Swedish Society for Industrial Flow and Heat Transfer, May 2026, Lund, Sweden
  Title: Towards efficient control of turbulence
- LKAB Company, February 2026, Kiruna, Sweden (held online)
  Title: Engineering applications of computational and data-driven fluid dynamics
- Swedish Society for Industrial Flow and Heat Transfer, May 2024, Finspång, Sweden
  Title: Artificial intelligence and data-driven algorithms for flow feature extraction and control
- Dutch OpenFOAM User group meetings, July 2023
  Title: Deep Reinforcement Learning in CFD, Implementation in OpenFOAM
- OpenFOAM Student Workshop at Delft University of Technology, January 2023
  Role: Invited OpenFOAM expert; provided guidance on advanced OpenFOAM applications for turbomachinery and answered questions from PhD researchers

Utilization of My Research

- During my postdoc, I developed the required numerical methods to study transients in hydraulic turbines and published the framework as an open-source package (`semiImplicitSlip`). The Research and Development department of Vattenfall, a leading energy production company in Europe, is now using my development to study the transient operation of their hydropower systems.
- I also have other open-source GitHub projects aimed at making my research beneficial for both the academic and industrial communities. For more information, please refer to my GitHub at github.com/salehisaeed.

Open-Source Software

- `TensorforceFoam`: A TensorFlow-based intrusive Deep Reinforcement Learning (DRL) framework for OpenFOAM. The Tensorforce package is utilized for the DRL computations. The term intrusive refers to the direct integration of the DRL agent within the CFD environment. Such integration eliminates the need for any external information exchange during DRL episodes. The framework is parallelized using the message passing interface (MPI) for Python (mpi4py) to manage parallel environments for computationally intensive CFD cases through distributed computing.
- `semiImplicitSlip`: A mesh motion library in OpenFOAM that implements a robust semi-implicit algorithm for slipping the mesh points on curved surfaces. The algorithm includes two steps, namely, an explicit step based on the general slip condition and an implicit step based on the Dirichlet condition. It employs the Laplacian smoothing equations to spread the mesh deformation into the domain. Additionally, a solid-body rotation may be added on top of the deformed mesh, which could be useful for modeling the runner region in transient operation of Kaplan turbines which contains simultaneous mesh deformation and solid-body rotation of the mesh.
- `Francis-99 transients`: Open-source case and code for CFD simulation of the transient operation of a Francis turbine in OpenFOAM.

Computer Skills

CFD: OpenFOAM, Fluent, CFX

Programming: C++, Python, MATLAB, LaTeX

AI / ML: TensorFlow, PyTorch

Post-processing: ParaView, Tecplot

General: Linux, Git

Languages

English: Fluent   Swedish: Fluent   Persian: Mother tongue